

An aerial photograph of a city skyline, likely Chicago, during sunset. The sky is filled with soft, orange and pink clouds. The city's skyscrapers are silhouetted against the bright horizon. In the foreground, the city's grid of buildings is visible. The text "Finding the Best Neighborhood to Live" is overlaid in a large, white, serif font. Below it, the names of the authors are listed in a smaller, white, sans-serif font. The overall mood is serene and urban.

Finding the Best Neighborhood to Live

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Meyer, and Nathan Smoocha

The current location finding tools are impersonal and insufficient

- People are relocating at lower rates, which may negatively impact the economy (Tavernise, 2019); (Kopf, 2018).
- “Top City” lists are impersonal, and too broad (Okulicz-Kozaryn, 2011).
- Tools like Teleport do take in current city to recommend new locations, but do not consider nuances between neighborhoods within a city.
- Tools like Niche force customers to search, rather than giving suggestions, and they do not offer data on essential services (like grocery stores).
- The current available tools lack interactive visualizations.

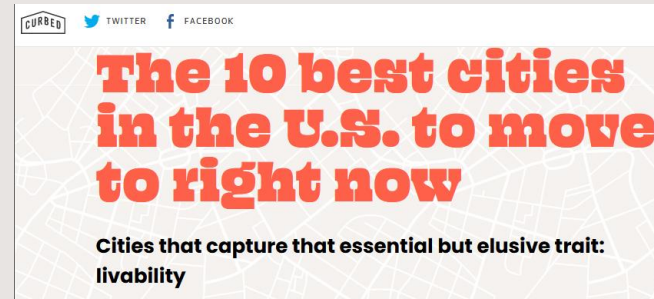


Figure 1. Web article title of best cities to move, Adapted from *The 10 best cities in the U.S. to move to right now*, 2020/2/11, Retrieved from <https://www.curbed.com/2020/2/11/21126341/best-cities-to-live-united-states>

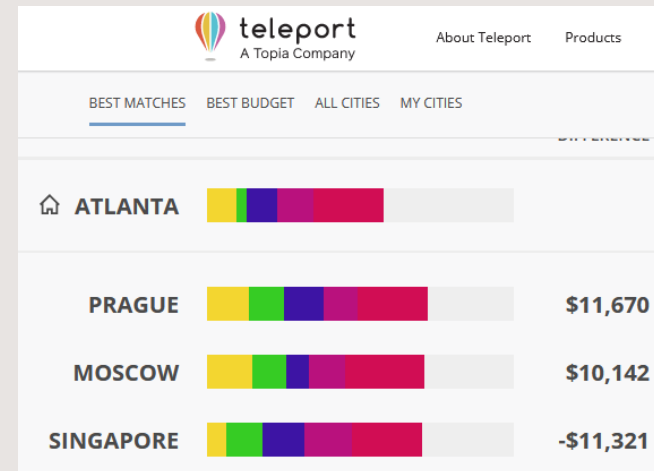


Figure 2. Teleport city recommendation, Adapted from *Teleport, MyCities*, Retrieved from <https://teleport.org/my-cities/>

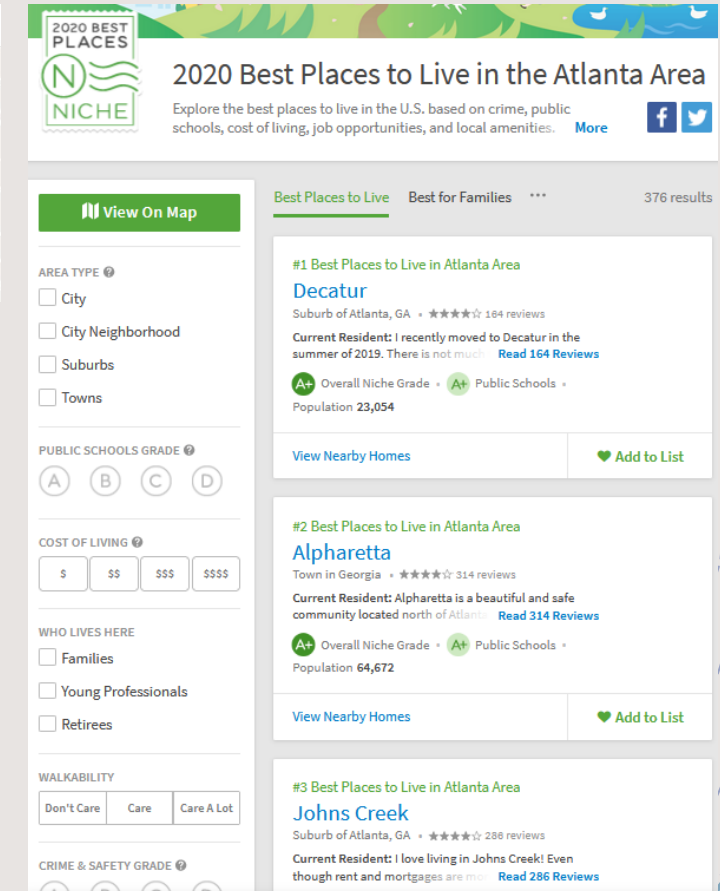


Figure 3. Places to live recommendation by niche.com, Adapted from *2020 Best Places to Live in the Atlanta Area*, Retrieved from <https://www.niche.com/places-to-live/search/best-places-to-live/m/atlanta-metro-area/>

We aim to create a data driven approach for people to find the best place to live based on places they already know they love

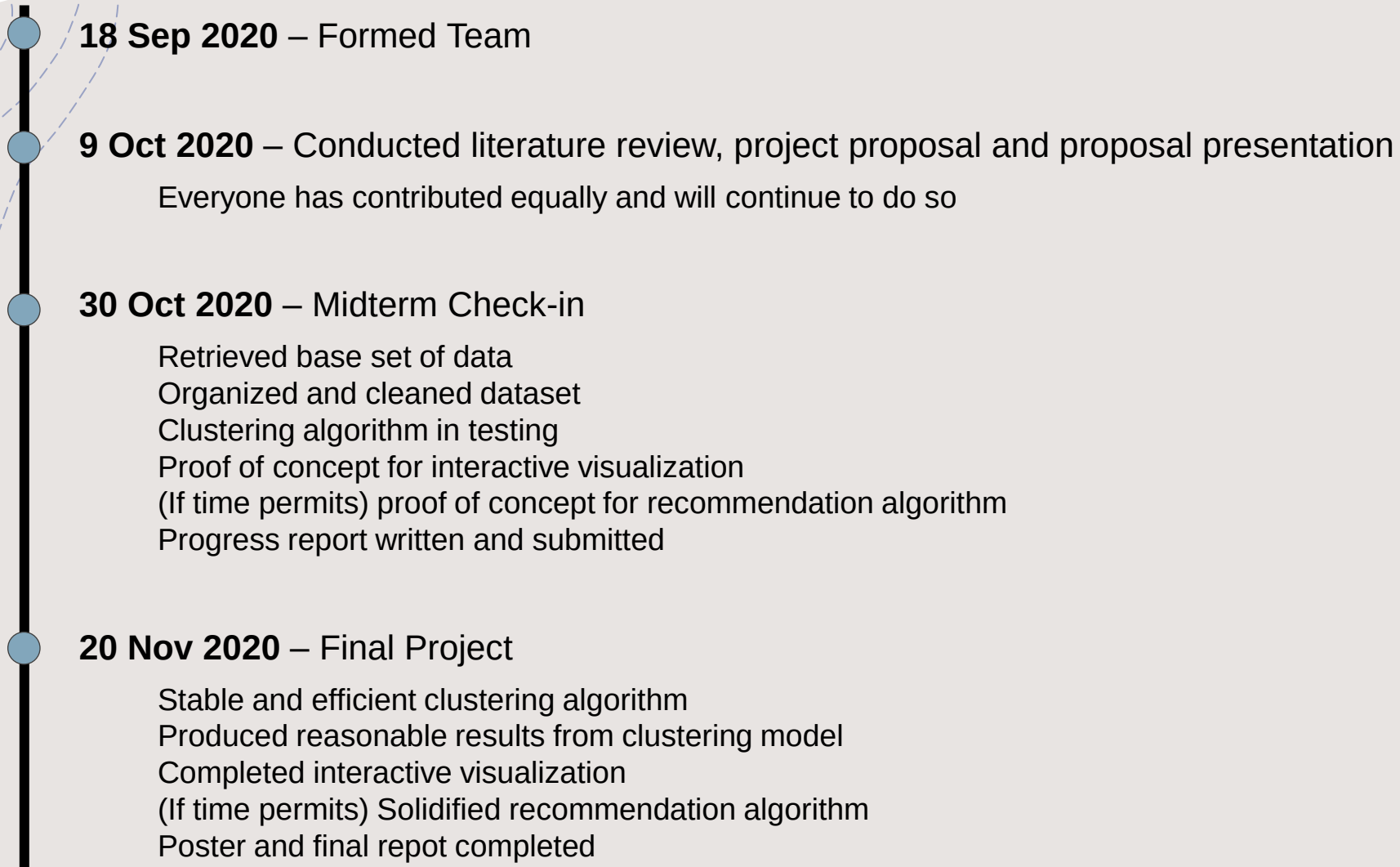
- The final tool will take a user inputted address, identify its census tract (i.e., neighborhood proxy), and provide a group of comparable neighborhoods from across the country based on similarity of attributes. Then, it will present the results in an interactive visual.
- Our project will utilize a richer dataset with new factors (e.g. essential services) and offer personalized results, neither of which are currently available.
- Risks include data quality/integrity, which could impact the results. However, the payoffs outweigh the risks. The biggest payoff is that the final tool will help ease people's apprehension about moving.
- Given free resources, the only cost is development time. We plan to utilize SQLite and Python. A visualization tool will be chosen later.
- Our project will be successful because we will be providing people a tool to help them make informative decisions about where to live.
- The success of our model will be based on volume, variety, and velocity. If the project were to launch in the future, then additional success measures would include user surveys, focus groups, and engagement metrics.

Solution



Figure 4. Wireframe of proposed neighborhood recommendation solution, Map Adapted from *Printable USA Blank Map*, Retrieved from <http://www.unitedstates-map.us/2018/07/printable-usa-blank-map-pdf.html>

We have determined key checkpoints to ensure completion of this project



Plan of Activities and Time Estimates, Per Group Member: data retrieval, organization, and cleaning (2 weeks: all team members); model construction, validation, and testing (2 weeks: Janicki, Meyer, Bell, & Smootha); and development of the interactive visualization (2 weeks: Dassanayake & Bodie).

We have completed a literature survey to assess the viability of this project

- We will use census tracts as proxies for neighborhoods
(U.S. Census Bureau, 1994)
- We looked at neighborhood factors people find important to consider for our project
(Ashkexari-Toussi et al., 2019); (Ayub et al., 2020); (Azadeh and Moghaddam, 2012); (He and Xia, 2020); (Okulicz-Kozaryn, 2011)
- We will cluster similar neighborhoods together
(Boongoen and Iam-On, 2018); (Fahad et al., 2014); (Hu and Pei, 2017)
- We reviewed ranking and recommendation models as other possible approaches
(Bobadilla et al., 2013); (Kim et al., 2020); (Kiseleva et al., 2015); (Liu and Yang, 2015); (Su et al., 2020); (Renjith et al., 2020)
- We also evaluated papers that employed a two-stage architecture, something that we could utilize with clustering and recommendation
(Xiaoyu et al., 2017); (Al-Ghossein et al., 2018); Wasid and Ali (2018)
- We looked at how various models were applied to geographical problems to understand potential challenges and approaches
(Nicholson et al., 2019); (Shashank and Schuurman, 2019); (Sadler et al., 2019)



Figure 5. Census tracts map image, Adapted from *Decision-Making Information Resources & Solutions* Warren Glimpse at ProximityOne, Retrieved from <https://proximityone.wordpress.com/2014/09/01/census-tract-block-group-block-geography/>

An aerial photograph of the Chicago skyline at sunset. The sun is low on the horizon, casting a warm, golden glow over the city. The sky is filled with soft, orange and pink clouds. The city's skyscrapers are silhouetted against the bright sky, with some buildings reflecting the sunset light. The Lake Michigan is visible on the left side of the image.

Appendix: Finding the Best Neighborhood to Live

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